

Chern-Simons Gauge Theories, Confinement, and the Chiral Spin Liquid

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We reconsider the problem of confinement in theories with fermions with a compact gauge symmetry in $2+1$ dimensions. We show that, as a result of the lack of gauge invariance of the determinant of the Dirac operator with a monopole background, the monopoles do not contribute to the Euclidean path integral. This result implies that fermions carrying the fundamental charge are not confined and that the spectrum of chiral-spin-liquid phases of theories of high-temperature superconductors supports separation of spin and charge.

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Compact quantum electrodynamics in $d=2+1$ dimensions (QED₃) provides a beautiful example on how non-perturbative effects can lead to the confinement of a fundamental quantum number such as the electric charge. This theory has (regular) monopole solutions^{1,2} which can be taken as instanton configurations in the evaluation of the generating functional path integral. Polyakov³ showed that the logarithmic interaction between electric charges becomes linear due to the monopole contribution. The theory is equivalent to the Coulomb gas which exhibits Debye screening of magnetic charges for arbitrarily weak coupling. The length scale of the crossover from logarithmic to linear potential becomes exponentially large at weak coupling. Polyakov further showed that screening of magnetic charge implies confinement of electric charge.

Anderson and co-workers⁴ have argued recently that gauge theories in $2+1$ dimensions may be relevant to the physics of high-temperature superconductors. In the “resonating-valence-bond” (RVB) picture of strong-coupling theories of high- T_c systems, a U(1) compact gauge field appears naturally. This U(1) gauge field represents the phase of the instantaneous overlap of the spin states on neighboring Cu atoms in the CuO₂ planes. The charge carriers (holes) move by flipping spins and their motion is affected by the gauge fields. As a result of the constraints imposed by the strong-coupling limit, the electron can be regarded as a bound state of a charged spinless boson (“holon”) and neutral spin- $\frac{1}{2}$ fermion (“spinon”). Holons and spinons have opposite unit U(1) charge. The RVB picture assumes the absence of long-range order among the spins (i.e., a spin liquid) and the overlaps between nearby spins are expected to fluctuate wildly. From the point of view of a gauge theory, this is the confinement regime for the U(1) charge. This means that holons and spinons cannot

separately be part of the physical spectrum *unless* somehow confinement is destroyed. However, this cannot be accomplished by a Higgs mechanism since, as it is well known,⁵ the spectra of confined and Higgs phases are smoothly connected if the matter fields carry the fundamental charge unless the symmetry is broken dynamically by condensates in the “adjoint” (charge 2). We conclude that a “true spin liquid,” in which there is no long-range order among its spins and which, at the same time, exhibits separation of spin and charge cannot correspond to a confined phase. In addition to a possible dynamical symmetry breaking, in $2+1$ dimensions we have the alternative of the spontaneous breakdown of time-reversal invariance. This is the chiral spin liquid (CSL) of Kalmeyer and Laughlin⁶ and Wen, Wilczek, and Zee⁷ (WWZ). The effective action for the RVB gauge fields in the CSL acquires an extra term,⁷ a Chern-Simons term. We will argue that, in this case, confinement is lost while the theory remains gauge invariant. As a result, this state supports true separation of spin and charge.

In this paper we reconsider the confinement problem in $2+1$ dimensions in compact gauge theories with spontaneous breakdown of time-reversal invariance. It has been argued by several authors⁸⁻¹⁰ that the situation becomes radically different when a Chern-Simons (CS) term is added to the originally massless U(1). It is well known¹¹ that in this case the photon becomes massive and the long-range correlations necessary for the Debye screening of the monopole gas are not present. Hence, monopoles and antimonopoles are necessarily bound to each other so tightly that they do not respond as a plasma but as a gas of “molecules” or “dipoles” which does not exhibit Debye screening. The Wilson loop for external charges no longer has an area-law behavior and confinement is destroyed. Although this deconfinement

scenario is physically sensible, actual calculations of the effects of the addition of a CS term to compact QED₃ rely on some unsatisfactory arguments already pointed out by the authors in Refs. 8–10. Since the CS term is purely imaginary in Euclidean space, classical solutions to the equations of motion become complex. More importantly, these complex configurations have infinite action.⁹ This problem can be circumvented if one considers the usual monopole solution and treats the CS term as a one-loop effect arising from the integration of charged fermions.¹⁰ Then, if one considers a one-monopole configuration, integration over a collective coordinate associated to “large gauge transformations” wipes out the monopole contribution. It is not clear, however, how to correctly generalize this argument to dilute multimonopole configurations.

We will discuss in this paper a different approach to this problem, based on a careful treatment of the gauge degrees of freedom when defining the generating functional for the model. We consider for simplicity a pure Abelian (compact) gauge theory coupled to charged massive fermions but, as we shall see, the non-Abelian case [say, an SO(3) theory with spontaneously broken symmetry and fermions in the fundamental representation] can be identically treated. Our arguments remain valid if instead of charged fermions one just considers the addition of a CS term.

Let us start from the three-dimensional Euclidean action,

$$Z = \int \mathcal{D}A_\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[\frac{1}{4} \int d^3x F_{\mu\nu}^2 + \int d^3x \bar{\psi} D[A] \psi \right], \quad (1)$$

where $D[A]$ is the covariant (Euclidean) Dirac operator with mass m . The fermions are taken in the fundamental representation of the gauge group. The mass m represents the spinon gap of the chiral spin liquid. It is important to stress that the gauge-field integration in Eq. (1) extends over all gauge-field configurations. One can make this explicit by means of the Faddeev-Popov procedure¹² which consists in writing the measure DA_μ in the form

$$DA_\mu = DA_\mu \delta[F[A^\omega]] \Delta_{\text{FP}}[A] D\omega, \quad (2)$$

where $A_\mu^\omega = A_\mu + (1/e)\partial_\mu\omega$ and Δ_{FP} is the Faddeev-Popov determinant associated with the $F[A]=0$ condition. Of course, for gauge-invariant models, insertion of Eq. (2) into Eq. (1) factors out a trivial ω integration after an appropriate change of variables. It is in this way that the integration measure for gauge-invariant theories becomes an integration over the space of fields restricted by the gauge symmetry (orbit space). The fermionic measure in Eq. (1) is not gauge invariant if monopole configurations are included. Thus, the ω integration is not trivial but it wipes out from Z contributions of any sector with magnetic charge $n \neq 0$.

Indeed, consider the fermionic measure $\mathcal{D}\mu_A[\psi]$ in Eq. (1). Under a gauge transformation this measure

changes by a Jacobian $J[A, \omega]$:

$$\mathcal{D}\mu_A[\psi^{-\omega}] = \mathcal{D}\mu_A[\psi] J[A, \omega]. \quad (3)$$

The value of the Jacobian J associated to this gauge transformation is

$$J[A, \omega] = \det D[A^\omega] / \det D[A]. \quad (4)$$

We see that if $J \neq 1$ (i.e., if one cannot define a gauge-invariant fermion determinant), the integration over ω does not trivially factorize. Indeed,

$$Z = \int DA_\mu \delta[F[A]] \Delta_{\text{FP}}[A] J[A, -\omega] \times D\omega D\bar{\psi} D\psi e^{-S[\bar{\psi}, \psi, A]}. \quad (5)$$

Let us evaluate the Jacobian of Eq. (4) for a monopole configuration $A_\mu^{(n)}$ with n units of magnetic charge. That is, with a flux $\Phi = 4\pi n/e$. Such a configuration can be realized by considering a magnetic field $B_\mu = \epsilon_{\mu\nu\alpha} \partial_\nu A_\alpha$ such that the Bianchi identity is violated,

$$\partial_\mu B_\mu = (4\pi n/e) \delta^{(3)}(\mathbf{x} - \mathbf{x}_a), \quad (6)$$

for a monopole located at $\mathbf{x} = \mathbf{x}_a$. Subtleties related to the definition of such a configuration, such as the appearance of Dirac strings, etc., will not be taken into account here since we have in mind the case of regular 't Hooft-Polyakov^{1,2} monopole solutions in the SO(3) model where these problems are absent.

It is well known^{11,13} that the three-dimensional fermion determinant is given by

$$\det D[A] = \exp \left[i \text{sgn}(m) \frac{e^2}{8\pi^2} \int d^3x \epsilon_{\mu\nu\alpha} A_\mu \partial_\nu A_\alpha + \frac{1}{24\pi|m|} \int d^3x F_{\mu\nu}^2 + O\left(\frac{\partial^2}{m^2}\right) \right], \quad (7)$$

where the first term on the right-hand side is the Chern-Simons action which will be called S_{CS} .

Now, for an n -monopole configuration S_{CS} is not gauge invariant. One can see this by writing

$$S_{\text{CS}}[A^{(n)\omega}] = S_{\text{CS}}[A^{(n)}] + \frac{e}{8\pi^2} \int d^3x B_\mu \partial_\mu \omega. \quad (8)$$

Integrating by parts the second term in Eq. (8) and dropping off a surface term, one gets¹⁴

$$S_{\text{CS}}[A^{(n)\omega}] - S_{\text{CS}}[A^{(n)}] = -n\omega(\mathbf{x}_a)/2\pi. \quad (9)$$

Using this result, one finally has for J ,

$$J[A^{(n)}, \omega] = e^{-in\omega(\mathbf{x}_a)/2}. \quad (10)$$

We can now perform the integration over ω in Eq. (5) to get

$$\int D\omega(x) e^{-in\omega(\mathbf{x}_a)/2} = \mathcal{N} \delta_{n,0}, \quad (11)$$

where $\mathcal{N}$ is a normalization constant.¹⁵

Hence, the ω integration wipes out all topologically nontrivial sectors from the path integral in Eq. (5) and the generating functional reduces to a path integral over the $n=0$ sector:

$$Z = \mathcal{N} \int DA_\mu^{(0)} \delta[F[A^{(0)}]] \Delta_{\text{FP}} D\bar{\psi} D\psi e^{-S[A^{(0)}, \bar{\psi}, \psi]}. \quad (12)$$

Thus, we have proven that the presence of charged fermions eliminates *any* monopole contribution from the QED₃ generating functional. In particular, even configurations of monopoles and antimonopoles with *total* magnetic charge zero are eliminated. In the usual argument,³ configurations of monopoles and antimonopoles are responsible for confinement. We can then conclude that when fermions are taken into account (or alternatively a CS term is included), since the configurations with monopoles are wiped out, the model is no longer confining. Interestingly enough, it is the contribution of gauge degrees of freedom, absent at the classical level, which eliminates (through the ω integration) topologically nontrivial sectors. This phenomenon very much resembles the one occurring in string theory¹⁶ or anomalous gauge theories^{5,17,18} where gauge degrees of freedom, classically absent, reappear at the quantum level if one correctly takes into account the anomaly (Weyl anomaly, gauge anomaly) within the gauge-fixing process. Also in the present case, a careful gauge fixing shows that gauge degrees of freedom do play a role at the quantum level: They eliminate from the generating functional, through the fermionic integration measure, all topologically nontrivial (i.e., monopole) sectors.

It is worthwhile to give yet another interpretation of our result. The Chern-Simons action, Eq. (8), is invariant under local gauge transformation only in the absence of monopoles since otherwise the Bianchi identity is violated. The gauge transformation $\omega(x)$ can be regarded as a field which restores gauge invariance in the same way as the phase of a scalar field preserves it in a Higgs theory. Indeed, the combination

$$\int d^3x \epsilon_{\mu\nu\lambda} (A^\mu + \partial^\mu \omega) \partial^\nu A^\lambda \quad (13)$$

is manifestly gauge invariant. Since Eqs. (9) and (14) differ only by an integration by parts, the field ω only gives a nonvanishing contribution where the Bianchi identity is violated by the monopoles. The phase field ω only appears in the presence of monopoles. It is natural to speculate that it is related to the bound states of the Dirac operator in 2+1 Euclidean dimensions.

We have discussed the Abelian case just to make clear the actual mechanism leading to the disappearance of confinement in QED₃ when fermions (or a CS term) are included. The extension to a non-Abelian model is straightforward. In fact, since non-Abelian regular monopoles are known to exist,^{1,2} problems associated to Abelian monopoles (in particular, handling Dirac strings) are avoided.

Consider as an example an SO(3) gauge theory spontaneously broken to U(1) by a triplet of Higgs fields. The classical equations of motion exhibit monopolelike solutions such that the U(1) electromagnetic-field tensor¹ corresponds to a monopole with magnetic flux $\Phi = 4\pi/e$. Multimonomopole ($n \neq 1$) solutions can also be constructed¹⁹ but their exact form will not be important

for our purposes. Instead, following Polyakov,³ we just consider a superposition of monopoles with magnetic charges $q_A = \pm 1$ fixed at points $\mathbf{x} = \mathbf{x}_A$ leading to a magnetic-charge- n configuration which approximates minima of the classical action provided monopoles are far from each other ($R_{AB} = |\mathbf{x}_A - \mathbf{x}_B| \gg e\eta$). The magnetic field $B_\mu^{(n)}$ in this case takes the asymptotic form

$$B_\mu^{(n)} \sim \sum_A \frac{q_A}{2} \frac{\mathbf{x} - \mathbf{x}_A}{|\mathbf{x} - \mathbf{x}_A|^3}. \quad (14)$$

We can now repeat the steps leading to the generating functional Eq. (12). To do so, we will have to evaluate the Jacobian J now in a non-Abelian background. We know^{11,13} that the fermion determinant leads again to a CS action which, in the present SO(3) case, takes the form

$$S_{CS}[A] = \frac{\epsilon^2}{4\pi} \int d^3x \epsilon_{\mu\nu\alpha} (F_{\mu\nu}^a A_\alpha^a - \frac{1}{3} \epsilon^{abc} A_\mu^a A_\nu^b A_\alpha^c). \quad (15)$$

As in the Abelian case, S_{CS} is not gauge invariant in a monopole background. Instead, one has an expression analogous to Eq. (8) with ω now taking values in the SO(3) algebra. Thus, instead of Eq. (12), we get

$$J[A, \omega] = \exp \left\{ i \sum_A \frac{q_A}{2} \omega(\mathbf{x} - \mathbf{x}_A) \right\} \quad (16)$$

and then the ω integration eliminates from Z all sectors with magnetic charge $n \neq 0$ exactly as in the Abelian case. Hence, the non-Abelian monopole contribution disappears from the partition function when fermions (or a CS term) are present so that confinement of external charges no longer holds in this case.

Even if the lack of gauge invariance of the fermion determinant is ignored, the presence of the induced Chern-Simons term significantly modifies the arguments presented by Polyakov³ proving confinement in QED₃. Indeed, let us briefly review how the multimonomopole contribution QED₃ was evaluated in Ref. 3 within the dilute-gas approximation.

Let us consider, as above, a superposition of widely separated monopoles, again located at points $\mathbf{x} = \mathbf{x}_A$ with magnetic charge $q_A = \pm 1$. The radius of each monopole core is $r = 1/e\eta$ and its magnetic field takes the form of Eq. (14). The action S_{cl} for such a multimonomopole configuration can be calculated by considering two types of contributions: that coming from each monopole core (region I) and that from a region II, far enough from each monopole. The first region can be visualized as being formed by spheres of radius R ($e\eta \ll R \ll R_{AB}$) centered at each monopole. Each sphere gives the same contribution to the total action. We call this contribution S^I . The second contribution to S_{cl} , S^{II} , comes from region II where an asymptotic form for the multimono-

pole magnetic field can be used. Hence,

$$S = S^I + S^{II} \\ = \sum_A \frac{\eta}{e} \epsilon \left(\frac{\lambda}{e^2} \right) + O \left(\frac{1}{e\eta R} \right) \\ + \frac{\pi}{2e^2} \sum_{A \neq B} q_A q_B \frac{1}{|\mathbf{x}_A - \mathbf{x}_B|} + O \left(\frac{1}{e^2 R} \right), \quad (17)$$

where $\epsilon(y)$ is a slowly varying function of y . Following Polyakov³ we conclude that a dilute gas of monopoles and antimonopoles behaves like a Coulomb gas, which exhibits Debye screening and where external charges are confined.

Consider now the case in which, in addition to the Yang-Mills Higgs action, one has a Chern-Simons term generated, for instance, by fermion integration. There is no finite-action monopolelike solution in this case. However, we can just consider the CS term as a one-loop effect and study the contribution of the pure Higgs-Yang-Mills monopole solution to the complete system. Now, although the CS action is zero for the (multimonopole) configurations, there is a new contribution to the effective action coming from terms linear in fluctuations (remember that the classical configuration is no longer an extremum of the complete action). Up to quadratic terms we find an action of the form

$$S = S^I + S^{II} + i \frac{e^2}{\pi} \int d^3x B_\mu^{(n)} a_\mu \\ + \frac{1}{2} \int d^3x f(x) S^{(2)}(x, y) f(y), \quad (18)$$

where f represents the fluctuations. After eliminating the linear term from Eq. (18) we obtain the effective action S

$$S = S^I + S^{II} \\ + i \frac{e^4}{32\pi^2} \int d^3x d^3y B_\mu^{(n)}(x) S_{\mu\nu}^{-1}(x, y) B_\nu^{(n)}(y) \\ + \frac{1}{2} \int d^3x f(x) S^{(2)}(x, y) f(y). \quad (19)$$

The new relevant contribution comes again from region II. Calling this new contribution S_{CS}^{II} , one finds after some calculations,

$$S_{CS}^{II} = - \frac{e^2}{16\pi^2} \sum_{A, B} q_A q_B R_{AB}. \quad (20)$$

Thus, the effective partition function Z_{MG} for the monopole gas is now

$$Z_{MG} = \sum_N \frac{\xi^N}{N!} \prod_{j=1}^N d\mathbf{x}_j \exp \left[- \frac{\pi}{2e^2} \sum_{A \neq B} \frac{q_A q_B}{R_{AB}} \right. \\ \left. + \frac{e^2}{16\pi^2} \sum_{A, B} q_A q_B R_{AB} \right]. \quad (21)$$

The presence of the linear-potential term in Eq. (21) suggests that confinement is destroyed: Monopoles and antimonopoles themselves may become confined. They may form a gas of "molecules," which does not screen external sources, instead of a charged plasma. Nelson²⁰

has argued that a Laplacian roughening model, closely related to the gas of Eq. (21), is in a plasma phase due to entropy effects. In our context, this would imply confinement. However, the coupling constant e in Eq. (21) scales like $(\text{length})^{-1/2} \sim |m|^{1/2}$. Thus, in the infrared, e becomes strong, as required by the assumption that the CSL is stable, and Nelson's arguments may not be applicable. An analogous problem was found in discussions of the role of hedgehogs in nonlinear σ models.^{21,22} This heuristic analysis also suggests that the monopoles do not contribute to the path integral. To conclude, in this paper we have shown that dynamical fermions coupled to gauge fields in 2+1 dimensions undo the confinement of charge by the gauge fields.

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¹⁵Equation (11) is also valid if, as in Ref. 10, one allows ω at infinity to be a nonzero constant while satisfying the Bianchi identity. This is the natural approach in the non-Abelian case which is discussed by Affleck *et al.* (Ref. 10).

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